

Name: _____

In-class Quiz #01: Numbering Systems

1. Convert the unsigned 8-bit binary numbers to decimal:
1) 0b11011011; 2) 0b00110011.

2. Convert the decimal numbers to unsigned 8-bit binary:
1) 62; 2) 98;

3. Convert the hexadecimal numbers (unsigned) to decimal:
1) 0xD9; 2) 0x2F;

4. Convert the decimal numbers to hexadecimal:
1) 1025; 2) 672.

5. Convert hexadecimal to binary and convert binary to hexadecimal
 - 1) 0b1011 0011 0101; 2) 0x8D0.

6. Convert the numbers into the specified number system (use 0-9 and A-J to represent the digit from 0 to 19. For example, you can use A to represent the digit 10, and use G to represent the digit 16.):
 - 1) Convert $(532)_6$ to base-8 (Radix = 8);
 - 2) Convert $(B9)_{12}$ to base-20 (Radix = 20).

7. Change the signed 8-bit binary, using two's complement representation, back to decimal:
 - 1) 0b11000100; 2) 0b11110001;

8. Convert the decimal numbers into signed binary using 2's complement representation, perform the addition, then convert the results back to decimal. (To add two signed 8-bit binary numbers using two's complement, simply align the values on the least significant bit and add, propagating any carry to the bit one position high. If you have a carry past the end of the word (overflow), just drop it.)
 - 1) $(-62)+89$; 2) $(-23)+(-38)$;

9. (extra, 10 pts) We know that overflow at the highest bit (sign bit) could happen when performing the additions of two signed binary numbers with 2's complement representation. But, overflow at the highest bit doesn't mean the result is wrong, for example, the following operation is correct.

$$\begin{array}{rcccccccc}
 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & & (-70) \\
 + & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & & (-31) \\
 \hline
 (1) & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & & (-101)
 \end{array}$$

Some people say we can record the carries happening at both the highest bit and the second highest bit to determine if the addition is correctly performed or not. For example

$$\begin{array}{rcccccccc}
 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & & (-70) \\
 + & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & & (-31) \\
 \hline
 (1) & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & & (-101) \\
 & \uparrow & \uparrow & & & & & & & & \\
 & C_1 & C_2 & & & & & & & &
 \end{array}$$

We have two variables C_1 and C_2 to record the carries. In the above example $C_1 = 1$ and $C_2 = 1$, for there carries at both the highest and the second highest bits. Please find if this method works (can find if the addition is correct or not). If it works, when ($C_1 = ?$; $C_2 = ?$) the addition operation is correct and when ($C_1 = ?$; $C_2 = ?$) the addition operation is wrong.